



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
27-06-2021

JEE(Main+Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **26** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PHYSICS

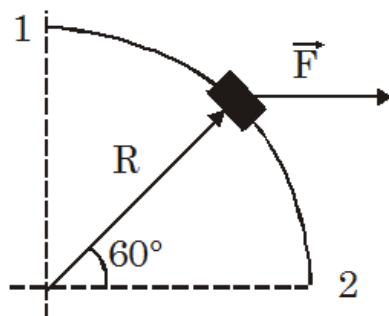
SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- A small wooden ball is falling from a great height. It attains a terminal speed of v_1 due to viscosity of air. When the same ball is released in a long column of liquid from the bottom, it rises up with a terminal speed of v_2 . If the temperature increases, select the correct statement(s) from the following. Neglect the change in any density due to increase in temperature.
 (A) v_1 increases (B) v_1 decreases (C) v_2 increases (D) v_2 decreases

Space for Rough Work

2. A small bead of mass $m = 0.15$ kg moves on a smooth wire bent in the horizontal plane in the form of an arc of a circle of radius = 50 cm (Fig. shows top view). At point 1, where the speed of the bead is $V = 7.5$ m/s, a constant horizontal force of magnitude 30 N starts acting on it as shown.



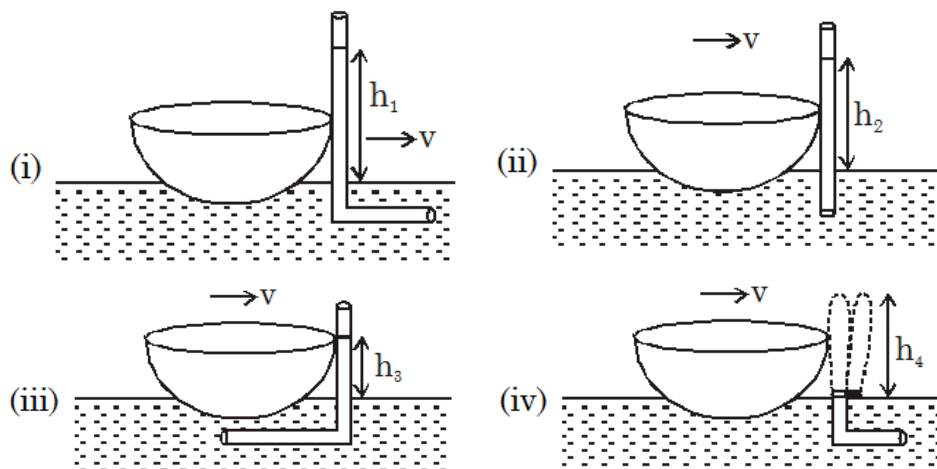
- (A) The speed of the bead at point 2 is $\frac{\sqrt{265}}{2}$ m/s.
- (B) The rate of change of speed when radius makes an angle of 45° with radius at 1 is $100\sqrt{2}$ m/s².
- (C) The normal force exerted by the wire at point 2 is approx 77 N.
- (D) The normal acceleration when radius makes an angle of 45° with radius at 1 is $100\sqrt{2}$ m/s².

Space for Rough Work

3. A small block of mass m is kept at the left edge of a plank of mass $2m$. The upper surface of the plank is rough ($\mu = 0.6$), but the surface in contact with the ground is smooth. Both the plank and block are made of same material. The length of plank is 10 m. The block is projected with a speed V_0 parallel to the length such that maximum heat is dissipated. The specific heat capacity of the material is $2000 \text{ J/kg } ^\circ\text{C}$. Assume that no heat is lost to the surroundings and the temperature of both the bodies remains same at all the times.
- (A) The maximum temperature change is $10^{-2} ^\circ\text{C}$.
- (B) If the maximum temperature change is $3.33 \times 10^{-3} ^\circ\text{C}$, the block stops at mid-point of the plank relative to the plank.
- (C) If the speed of projection is $v_0/2$, the temperature will change by $2.5 \times 10^{-3} ^\circ\text{C}$.
- (D) When the temperature change is maximum, the entire kinetic energy as seen from ground frame is dissipated in the form of heat.
4. A polytropic process $P/V = \text{constant}$ is carried out on 20 moles of monoatomic gas enclosed in a vertical cylinder. The cylinder is enclosed by a light piston of area 20 cm^2 at the top. The piston and walls of the cylinder are thermally insulated but the bottom is thermally conducting having $k = 2 \text{ W/mK}$. Its thickness is 2mm and it is kept on a reservoir maintained at a constant temperature of 400 K. The initial temperature of the gas is 300 K. Atmospheric pressure = 100 kPa.
- (A) The piston needs to be pushed downwards with a constant force to carry out the process.
- (B) The volume of gas increases with time as $v = \sqrt{\frac{1}{3} - \frac{1}{12} e^{-\frac{t}{300}}} \text{ m}^3$.
- (C) The temperature of the gas increases with time as $T = 100 \left(4 - e^{-\frac{3t}{500}} \right) \text{ K}$.
- (D) The force exerted by external agent on the piston to ensure this process is 200 N.

Space for Rough Work

5. From a boat moving at a speed $v = 18 \text{ km/h}$ in stationary water, we immersed a uniform tube as seen in the four cases below. The figures are merely representative and heights of water in the tube can be more or less than the figures shown. In case (iv) the open end of the tube is at the level of free surface of water. In that case, h_4 represents the height to which water jet will reach.



- (A) $h_1 = h_4 > h_2 > h_3$
 (B) $h_1 > h_4 > h_2 > h_3$
 (C) $h_2 = h_3$
 (D) height in all the cases is same and equal to 1.25 m.

Space for Rough Work

6. A cuboidal container open at the top with square shaped base of mass 2 kg has inner dimensions as $10\text{ cm} \times 10\text{ cm} \times 20\text{ cm}$. It is filled with one litre water and kept on a rough horizontal surface $\mu = 0.2$. A gradually increasing force is applied on the container in horizontal direction perpendicular to the face. Assume that the container does not topple at any time. Density of water is 1 gm/cc .
- (A) The container starts sliding when the force is 6 N.
- (B) When the force is 36 N, the angle made by the free water surface with the horizontal is 45° .
- (C) When the force is 46 N, the angle made by the free water surface with the horizontal is 53° .
- (D) The force at which water starts spilling out is 66 N.
7. An electric engine pulls the train having a total mass of 2000 ton. Assuming that the net power of the electric engine is constant at 1800 kW, and the coefficient of friction is 0.005.
- (A) The acceleration of the train at the moments when its speed u is 4 m/s is 0.175 m/s^2 .
- (B) The acceleration of the train at the moments when its speed u is 3 m/s is 0.65 m/s^2 .
- (C) The maximum velocity of the train is 18 m/s.
- (D) For the given condition to be true, the initial speed should be at least 0.18 m/s.

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 1 - 3

In a container fitted with a piston, we have one mole of a gas or a gaseous mixture whose total quantity is one mole. The composition of the gas is indicated in column-1. In column-2, the equation of the process carried out on the gas is mentioned. The column-3 mentions the pressure of the gas at the end of the process. The initial state of the gas in all the cases is pressure $P_0 = 10^5$ Pa and volume $V_0 = 25$ litre. Take $R = 25/3$ J/mol K.

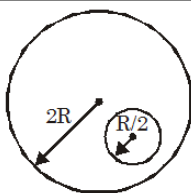
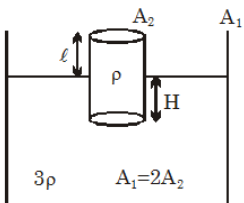
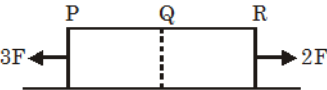
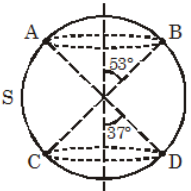
	Column-I		Column-II		Column-III
(I)	Monoatomic gas	(i)	$P = \text{Constant}$	(P)	$P = 10^5$ Pa
(II)	Diatomic gas	(ii)	$V = \text{constant}$	(Q)	$P = 2 \times 10^5$ Pa
(III)	Mixture of monoatomic gas and diatomic gas in molar ratio 1:1	(iii)	$PV^{1.5} = \text{constant}$	(R)	$P = 0.5 \times 10^5$ Pa
(IV)	Non linear polyatomic gas	(iv)	$PV^{-1} = \text{constant}$	(S)	$P = 4 \times 10^5$ Pa

- In which of the following combinations, heat is given to the gas but its temperature is decreasing?
 (A) (I) (i) (P) (B) (I) (iii) (R) (C) (II) (iv) (R) (D) (III) (iv) (R)
- In which of the following combinations, no heat is given to the gas or extracted from it?
 (A) (I) (iii) (Q) (B) (II) (iii) (S) (C) (III) (iii) (Q) (D) (IV) (iv) (Q)
- In which of the following combinations, the temperature of the gas increases?
 (A) (I) (i) (P) (B) (II) (ii) (R) (C) (III) (iii) (R) (D) (IV) (iv) (S)

Space for Rough Work

Paragraph for Questions 4 - 6

Column-I shows a system for which two numbers x and y are defined, column-II shows value of x and column-III shows the value of y .

Column-I	Column-II	Column-III
 (I) There is a small bubble inside a bigger bubble. Surface tension of bigger bubble is S and of smaller bubble is $2S$. Pressure outside bigger bubble is 0. Pressure inside smaller bubble $= xS/R$. Both bubbles have monoatomic gas in them $y = \frac{\text{no. of moles in between bigger \& smaller bubble}}{\text{no. of moles in smaller bubble}}$	(i) 2	(P) 3
 (II) A uniform solid cylinder of height h and density ρ is floating in a liquid of density 3ρ as shown. $x = \frac{l}{H}$ If cylinder is pushed down by a distance $\frac{H}{y}$, it is completely submerged in water.	(ii) 6	(Q) 7
 (III) Rod PQR is of uniform linear mass density and cross section A. Young modulus of rod is σ. In natural state $PQ = QR = \frac{l}{2}$; Elongation in part $PQ = \frac{xFl}{8A\sigma}$; Elongation in part $QR = \frac{9Fl}{A\sigma y}$	(iii) 18	(R) 5
 (IV) We have a spherical bubble of radius R, made up by liquid having surface tension S. AB and CD are two parallel planes as shown in figure. Net force of surface tension on part $ABCD$ due to rest of the part is $\alpha \pi R S / \beta$. There is no common factor in α and β. x defined as $x = 3\alpha/14$, y is defined as $y = \beta/5$	(iv) 11	(S) 8

4. Which of the following is correctly matched :

- (A) (IV) (i) (R) (B) (IV) (ii) (S) (C) (I) (iii) (Q) (D) (III) (ii) (P)

5. Which of the following is correctly matched :

- (A) (I) (iv)(P) (B) (I) (iii) (R) (C) (II) (i) (P) (D) (IV) (i) (S)

6. Which of the following is correctly matched :

- (A) (I) (i) (Q) (B) (IV) (ii) (R) (C) (III) (ii) (S) (D) (I) (iii) (S)

Space for Rough Work

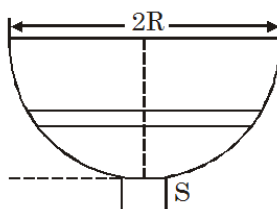
SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

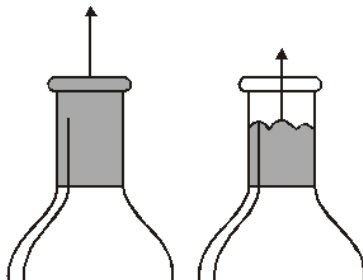
1. A hemispherical vessel has a small orifice at its bottom. It is filled up to the top with a liquid. Find the time taken (in sec) for emptying the vessel through the orifice at the bottom. R is the radius of the bowl and orifice has radius of $\frac{7}{3}$ mm. Take $R = 20$ cm. Assume that the surface of the water always has a much smaller velocity than that of the water coming from the orifice. Fill $\frac{t}{100}$ (in sec) round off to nearest integer.



2. A man of weight W hangs at the end of a light extensible rope for which the product of force constant and its natural length is $(2W)$ Newton, the other end being fastened to a fixed point. He proceeds to climb up the rope to reach the fixed point. If he were to climb the same distance up an inextensible rope, he would have to do a work of 600 J. What is the work W' (in J) he has to do in this case? Fill $W'/100$.

Space for Rough Work

3. In order to pull the entire cork from the bottle (Figure), it is necessary to perform work of 8J. What work (in J) is needed to uncork the bottle, which has broken off and the upper half of the cork has broken off? Cork is weightless.



4. A 1 m^2 metal plate glides on top of an 1 mm thick oil layer with constant velocity along a sloping plane. The weight of the metal plate 100 N and the dynamic viscosity of the oil is $0.1 \text{ Pa}\cdot\text{s}$. Determine the velocity (in cm/s) of the metal plate if the sloping angle of the plane is 5° . Disregard all end effects around the plate. Assume laminar flow and that the velocity profile in the oil is linear. Round off to nearest integer.
5. Container filled with air under pressure P_0 contains a soap bubble of radius R . The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S , P_0 is found to be $\frac{16nS}{R}$ SI unit Find n .

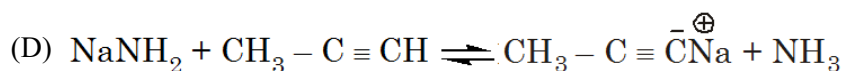
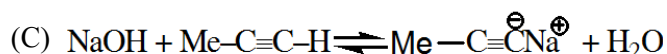
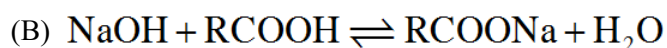
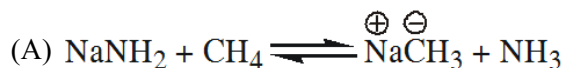
Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 28)

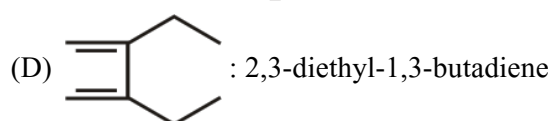
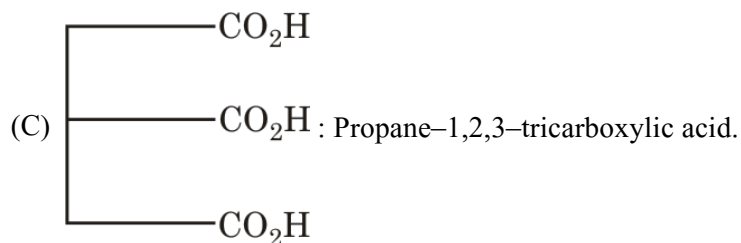
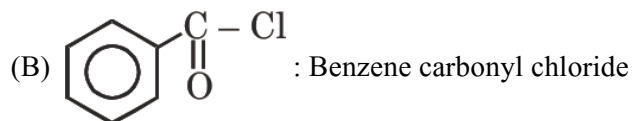
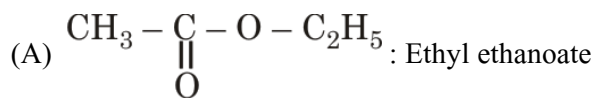
- This section contains **SEVEN (07)** questions.
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following reaction equilibrium is favoured in forward direction.



Space for Rough Work

2. Which of the following statement is correct regarding IUPAC name ?



3. Barium permanganate solution (20ml, 0.1M) is mixed with 0.1N I^- , giving precipitate of IO_3^- and MnO_2 . Resulting solution is filtered & titrated against Mo^{3+} , giving MoO_2^{2+} and Mn^{2+} , which required 0.5 M, 10ml acidified Mo^{3+} . Select the correct option(s)

(A) Volume of I^- solution taken is 30 ml

(B) Volume of I^- solution taken is 50 ml

(C) Per mole Mn^{2+} formed, 4 moles of H^+ are consumed

(D) Per mole IO_3^- formed, 2 moles of MnO_4^- are consumed

Space for Rough Work

4. Gases A & B are found in atmosphere in quantities of 200 and 500 per trillion (10^{12}) by volume.
($R = 0.08 \text{ atm lit/mole.K}$)
- (A) In mid -latitude troposphere (27°C and 1 atm) molarity of A is 8.33×10^{-12}
- (B) In antarctic stratosphere (27°C & 0.5 atm) molarity of B will 1.04×10^{-11}
- (C) In antarctic stratosphere (27°C & 0.5 atm) molarity of A will 7.05×10^{-11}
- (D) In mid -latitude troposphere (27°C and 1 atm) molarity of B is 4.39×10^{-12}
5. Hydrazine (N_2H_4) undergoes following reactions when heated at 1000 K as
 $\text{N}_2\text{H}_4(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) ; K_{p1}$
 $\text{N}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) ; K_{p2}$
 In a rigid vessel $\text{N}_2\text{H}_4(\text{g})$ was introduced at 1 atm initial pressure at 1000K . Total pressure at equilibrium was found to be 1.4 atm and partial pressure of H_2 was 0.3 atm .
 Select the correct option(s)
- (A) Partial pressure of $\text{N}_2\text{H}_4(\text{g})$ at equilibrium is 0.7 atm
- (B) P_{NH_3} at equilibrium is 0.2 atm
- (C) $K_{p1} = \frac{6}{7} \times 10^{-1} (\text{atm})^2$
- (D) $K_{p2} = \frac{4}{21}$

Space for Rough Work

6. The formal charges on different atoms in the Lewis dot structures of N_3^- are
(A) $-1, +1, -1$ (B) $-1, +1, 0$ (C) $-2, +1, 0$ (D) $0, +1, -2$
7. Which of the following specie(s) is/are having cyclic structure :-
(A) $\text{B}_3\text{O}_6^{3-}$ (B) $\text{B}_3\text{N}_3\text{H}_6$ (C) $\text{P}_4\text{O}_{12}^{4-}$ (D) S_3O_9
-

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

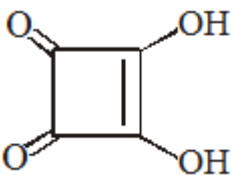
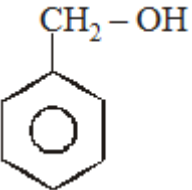
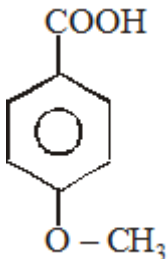
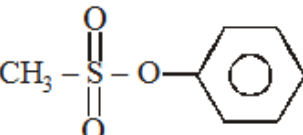
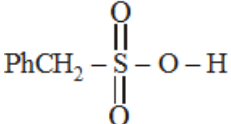
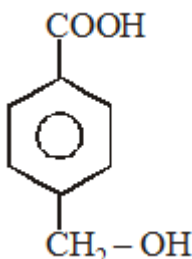
Paragraph for Questions 1 - 3

Column - I	Column - II	Column - III
(P) Orthoboric acid	(1) Intermolecular hydrogen bonding	(i) Slippery is nature
(Q) Ortho nitrophenol	(2) Intra molecular hydrogen bonding	(ii) Cage like structure
(R) Carbon monoxide	(3) Dipole-dipole interaction	(iii) Gas at normal temp.
(S) Ice	(4) Co-ordinate bond	(iv) Solid at normal temp.

- Which of the following is not related to 'P'.
 (A) 2 (B) i (C) iv (D) 3
- Which of the following matched is correct
 (A) Q – 2 – ii (B) R – 4 – iii (C) S – 1 – i (D) S – 4 – ii
- Which of the following melting point order is incorrect
 (A) $P > R$ (B) $P > S$ (C) $Q > R$ (D) $R > S$

Space for Rough Work

Paragraph for Questions 4 - 6

Coloumn-I	Coloumn-II	Coloumn-III
(P) 	(A) 	(w) Aromatic
(Q) 	(B) NaHCO_3	(x) Anti aromatic
(R) 	(C) 	(y) Functional isomers
(S) 	(D) 	(z) CO_2 gas evolve

4. Correct match regarding (P)

- (A) D, y, x (B) B, z, w (C) C, w, y (D) A, w, x

5. Correct match regarding (R) is

- (A) D, B, z, y (B) B, z, C, y (C) x, B, D (D) A, y

6. Correct match regarding(s) is

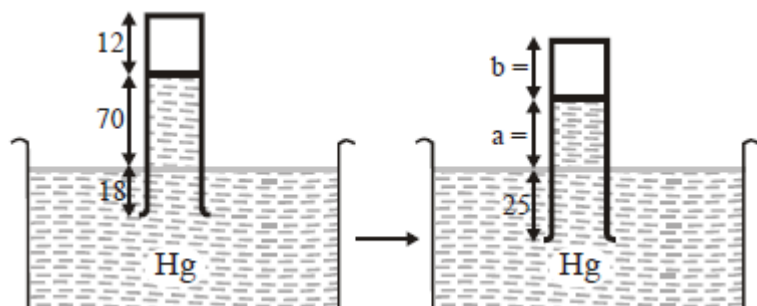
- (A) A, y, x (B) D, y, z (C) x, c (D) C, y, B, z

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:
Full Marks : +3 If ONLY the correct answer is chosen.
Zero Marks : 0 In all other cases.

1. How many structural isomers are possible for molecular formula $C_3H_4Cl_2$.
2. A H-like species is excited (may be from ground state or excited state) by a photon wavelength λ . When electrons return back to ground state photons of six different wavelength are observed. Out of six '3' photons have wavelength less than λ . Find the 'n' of energy level in which electrons are present initially.
3. An ideal gas was trapped inside an inverted tube of length 100 cm filled with Hg as shown in figure. What is the length of air column when tube is pushed down by 7 cm under isothermal conditions.



Space for Rough Work

4. The dissociation equilibrium of a gas AB_2 can be represented as $2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$
The degree of dissociation is α and it is small compared to 1. The expression relating the degree of dissociation (α) with equilibrium constant K_p and total pressure P is $(2K_p/P)^{\frac{1}{x}}$. Then x will be :
5. Find out the number of pairs in which boiling point of first compound is less than second?
($B(Me)_3$, BF_3) ; ($SiCl_4$, CCl_4) ; (H_2 , O_2) ; (NF_3 , $N(Me)_3$) ; (H_3BO_3 , $B(OMe)_3$) ; (CO , N_2) ; (H_2O , HF)
-

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $a_i, i = 1, 2, \dots, 9$ are squares of odd integers, then $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is always a multiple of
 (A) 4 (B) 16 (C) 64 (D) 32
2. Let the area Δ and its one of angle θ be given, when the side opposite to the given angle is minimum, then the length of remaining two sides are $\sqrt{\frac{\lambda\Delta}{\sin\theta}}, \sqrt{\frac{\mu\Delta}{\sin\theta}}$, then-
 (A) $\lambda + \mu = 4$ (B) $\lambda^2 + \mu^2 = 8$ (C) $\lambda + \mu = 2$ (D) $\lambda^2 + \mu^2 = 2$

Space for Rough Work

3. Let $P(x) = x^{32} - x^{25} + x^{18} - x^{11} + x^4 - x^3 + 1$. Which of the following are correct?
- (A) Number of real roots of $P(x) = 0$ is zero.
 (B) Number of imaginary roots of $P(x) = 0$ is 32.
 (C) Number of negative roots of $P(x) = 0$ is zero.
 (D) Number of imaginary roots of $P(x) + P(-x) = 0$ is 32.
4. Sum of infinite series $1 + \left(1 + \frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(1 + \frac{1}{2} + \frac{1}{2^2}\right) \left(\frac{1}{3^2}\right) + \dots$ is less than or equal to-
- (A) $\frac{9}{5}$ (B) 10 (C) $\frac{100}{7}$ (D) 1
5. In $\triangle ABC$, $\sin A \sin B \sin C = \frac{3 + \sqrt{3}}{8}$ and $\cos A \cos B \cos C = \frac{\sqrt{3} - 1}{8}$, then an angle of the triangle can be
- (A) 30° (B) 45° (C) 60° (D) 75°

Space for Rough Work

6. For the equation $2 \cos 2x \left(\cos 2x - \cos \left(\frac{2014\pi^2}{x} \right) \right) = \cos 4x - 1$; which of the following is/are correct ?
- (A) The number of solutions of the equation is 3.
- (B) The sum of all the positive solutions is 1080π
- (C) The number of positive solutions is 4
- (D) The sum of all positive solutions is 1008π
7. Consider a fixed point $P(5, 2)$ and also Q & R be two distinct variable points moving on $y - x = 0$ and $y = 0$ respectively, such that $|PQ + PR + QR|$ is minimum then
- (A) coordinate of R will be $\left(\frac{29}{5}, 0\right)$
- (B) coordinate of Q will be $\left(\frac{29}{10}, \frac{29}{10}\right)$
- (C) coordinate of R will be $\left(\frac{29}{7}, 0\right)$
- (D) minimum value of $|PQ + QR + PR|$ is $\sqrt{58}$

Space for Rough Work

SEC-2 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 1 - 3

Let A(2, 3) and B(5, 9) be two given points. If P is a variable point such that $\frac{PA}{PB} = K$ ($K > 0, K \neq 1$) then locus of P gives a circle. Column-I gives ratio $\frac{PA}{PB}$, Column-II gives center of circle described by locus of P and Column-III gives its corresponding radius.

Column-I		Column-II		Column-III	
(I)	$\frac{PA}{PB} = 2$	(i)	(1, 1)	(P)	$2\sqrt{5}$
(II)	$\frac{PA}{PB} = \frac{1}{2}$	(ii)	$\left(\frac{43}{8}, \frac{78}{8}\right)$	(Q)	$\frac{\sqrt{5}}{2}$
(III)	$\frac{PA}{PB} = 3$	(iii)	(6, 11)	(R)	$\frac{9\sqrt{5}}{8}$
(IV)	$\frac{PA}{PB} = \frac{1}{3}$	(iv)	$\left(\frac{13}{8}, \frac{18}{8}\right)$	(S)	$\frac{8\sqrt{5}}{9}$

- Which of the following is the only **CORRECT** combination ?
 (A) (I) (ii) (Q) (B) (II) (i) (P) (C) (III) (iv) (S) (D) (I) (iii) (R)
- Which of the following is the only **CORRECT** combination ?
 (A) (I) (iv) (P) (B) (III) (iv) (S) (C) (I) (iii) (P) (D) (IV) (iv) (S)
- Which of the following is the only **INCORRECT** combination ?
 (A) (I) (iii) (P) (B) (II) (i) (P) (C) (III) (ii) (R) (D) (IV) (iv) (S)

Space for Rough Work

Paragraph for Questions 4 - 6

The parabola $y^2 = 4ax$ has a chord AB joining points $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$. Column-I represents nature of chord AB, Column-II represents relation between t_2 and t_1 and Column-III represents locus of mid point of AB.

Column-I		Column-II		Column-III	
(I)	AB is normal chord	(i)	$t_2 = -t_1 - \frac{1}{2}$	(P)	$y^2 = 2a(x - a)$
(II)	AB is focal chord	(ii)	$t_2 = -\frac{4}{t_1}$	(Q)	$y = 2a$
(III)	AB subtends 90° at $(0, 0)$	(iii)	$t_2 = -\frac{1}{t_1}$	(R)	$y^4 - 2a(x - 2a)y^2 + 8a^4 = 0$
(IV)	AB is inclined at 45° to axis of parabola	(iv)	$t_2 = -t_1 - \frac{2}{t_1}$	(S)	$y^2 = 2a(x - 4a)$

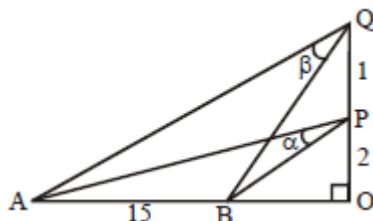
4. Which of the following is the only **CORRECT** combination ?
 (A) (I) (iv) (R) (B) (II) (iii) (S) (C) (III) (ii) (P) (D) (IV) (i) (S)
5. Which of the following is the only **CORRECT** combination ?
 (A) (I) (ii) (Q) (B) (II) (iii) (R) (C) (III) (ii) (S) (D) (IV) (i) (P)
6. Which of the following is the only **INCORRECT** combination ?
 (A) (I) (iv) (R) (B) (II) (iii) (P) (C) (III) (ii) (S) (D) (IV) (i) (R)

Space for Rough Work

SEC-3 : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories:
Full Marks : +3 If ONLY the correct answer is chosen.
Zero Marks : 0 In all other cases.

1. In the diagram shows $AB = 15$, $OP = 2$, $PQ = 1$, $\angle APB = \alpha$, $\angle AQB = \beta$, $\angle AOQ = 90^\circ$, then the value of $\frac{1}{3 \cot \beta - 2 \cot \alpha}$ is



2. Consider $f(x, y) = |x - 1| + 2|y - 2| + |xy|$. Let S be solution set of all possible ordered pairs (x, y) for which $f(x, y) = f(y, x)$. If (α, β) is an element of set S, $\alpha \neq \beta$, then maximum possible value of $\alpha + \beta$ is
3. Let $A_1, A_2, A_3, \dots, A_{11}$ be 11 arithmetic means and $G_1, G_2, G_3, \dots, G_{11}$ be 11 geometric means and $H_1, H_2, H_3, \dots, H_{11}$ be 11 harmonic means between two positive numbers a and b. If $ab = 9$ and value of $\prod_{k=1}^5 (A_k \cdot G_{12-2k} \cdot H_{12-k})$ is p^q where p is a prime number and q is a natural number then value of $\frac{q}{p}$ is

Space for Rough Work

4. The number of possible ordered pairs (x, y) (where x and y are integers) which satisfy $\frac{3}{x} + \frac{5}{y} = \frac{21}{xy} + 1$ is
5. The number of solutions of the simultaneous equations $2\sin^2 x + \sin^2 2x = 2$, $\sin 2x + \cos 2x = \tan x$ in $[0, 4\pi]$ satisfying the condition $2\cos^2 x + \sin x \leq 2$ is
-

Space for Rough Work

Space for Rough Work